MODULE 27 · DOMAIN 2 - PROCESS

Earned value and forecasting

Planned value, earned value and actual cost; variances and indices; forecasting the final cost; and reading the S-curve.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 27 of 49

admin@sthithapragna.com

For the time spent and money spent, **how much work is actually done?**

admin@sthithapragna.com

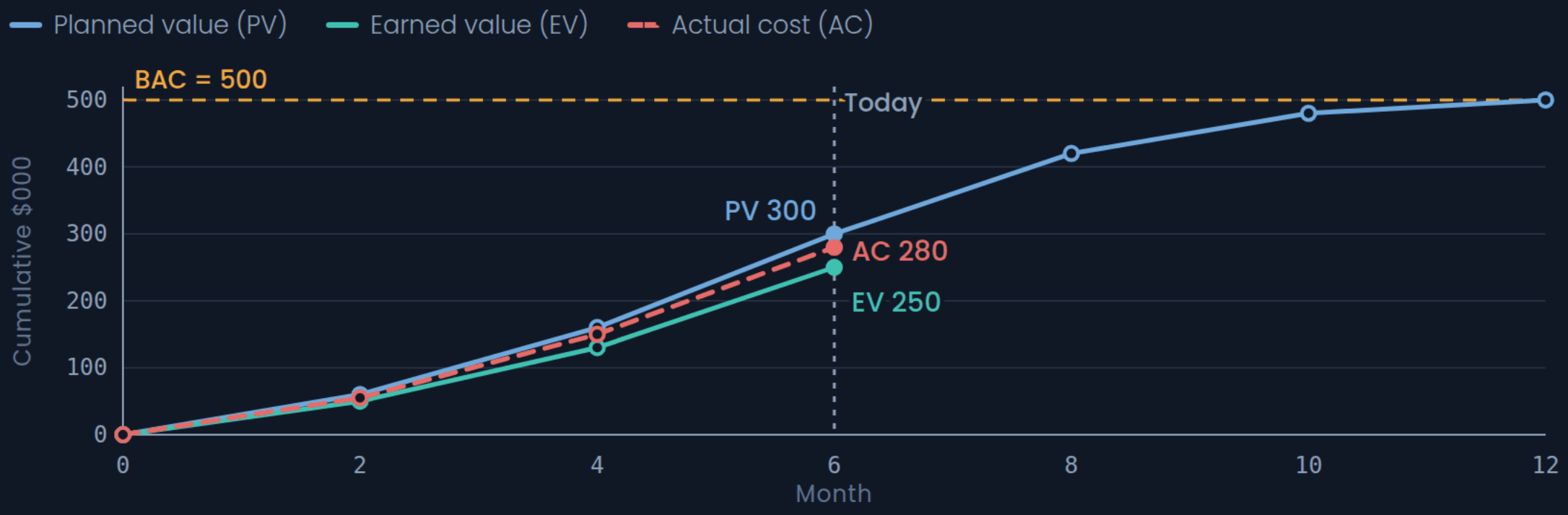
Four numbers to start with

TERM	MEANING	QUESTION IT ANSWERS
BAC, budget at completion	The total planned budget for the work	What should the whole thing cost?
PV, planned value	The budgeted value of work planned by now	What should be done by now?
EV, earned value	The budgeted value of work actually done	What is actually done?
AC, actual cost	What has actually been spent to date	What did it cost?

ON THE EXAM

EV and PV are measured in the budget's money, not in hours or percentages. That is what makes them comparable.

Reading the earned value S-curve



ON THE EXAM

Earned value below planned: behind schedule. Actual cost above earned: over budget.

Variances and indices

MEASURE	FORMULA	BAD NEWS WHEN
Cost variance (CV)	EV - AC	Negative: over budget
Schedule variance (SV)	EV - PV	Negative: behind schedule
Cost performance index (CPI)	EV / AC	Below 1: over budget
Schedule performance index (SPI)	EV / PV	Below 1: behind schedule

ON THE EXAM

Every formula starts with EV. Variance subtracts; index divides. Negative, or below 1, is bad.

The chart's numbers, worked through

- CV = 250 − 280 = **−30**. Over budget by $30,000.
- SV = 250 − 300 = **−50**. Behind schedule by $50,000 of work.
- CPI = 250 / 280 = **0.89**. Getting about 89 cents of work for each dollar.
- SPI = 250 / 300 = **0.83**. Progressing at about 83 percent of the planned rate.

> ON THE EXAM
>
> Schedule variance is measured in money: $50,000 of planned work not yet done.

Interpreting the indices

INDEX	ABOVE 1	EQUAL TO 1	BELOW 1
CPI	Under budget	On budget	Over budget
SPI	Ahead of schedule	On schedule	Behind schedule

ON THE EXAM

Both below 1: over budget and behind. The worst combination, and the most common exam answer.

Forecasting the final cost

FORECAST	FORMULA	USE WHEN
EAC, typical	BAC / CPI	Current cost performance will continue
EAC, atypical	AC + (BAC - EV)	The variance was a one-off; the rest goes to plan
EAC, cost and schedule	AC + (BAC - EV) / (CPI x SPI)	Schedule pressure will also affect cost
EAC, new estimate	AC + bottom-up ETC	The original plan is no longer valid
ETC	EAC - AC	How much more will be spent
VAC	BAC - EAC	Negative: forecast to finish over budget

ON THE EXAM

The question's wording chooses the EAC formula. 'Will continue' or 'typical': BAC / CPI.

Forecasting, worked through

- Typical: EAC = 500 / 0.893 = **560**. Expect to finish $60,000 over budget.
- ETC = 560 - 280 = **280** still to spend. VAC = 500 - 560 = **-60**.
- Atypical, if the overrun was a one-off: EAC = 280 + (500 - 250) = **530**.
- If schedule pressure also drives cost: 280 + 250 / (0.893 x 0.833) = about **616**.

> ON THE EXAM
>
> Same project, three forecasts. The story in the question decides which one is right.

To-complete performance index

- **TCPI = (BAC - EV) / (BAC - AC)**: work remaining divided by money remaining.
- Chart's project: (500 - 250) / (500 - 280) = 250 / 220 = **1.14**.
- The team must deliver $1.14 of work for every remaining dollar to finish on budget.
- A TCPI well above 1, with a CPI below 1, means finishing on budget is unrealistic.
- If a new EAC is approved, use (BAC - EV) / (EAC - AC) instead.

> ON THE EXAM
>
> TCPI above 1 means the future must be more efficient than the plan. Compare it with the current CPI.

What earned value does not tell you

- SPI drifts towards 1 near the end, because EV eventually equals PV, even on a late project.
- It does not show the **critical path**: good SPI can hide a late critical activity.
- It does not measure **quality** or **value**: work can be 'earned' and still be wrong.
- It relies on honest **progress rules**, such as 0/100 or 50/50, agreed in advance.

> ON THE EXAM
>
> Use earned value with critical path and quality checks, never alone.

Earned value on agile work

- A **release burnup** chart plays the same role: planned scope versus completed scope over time.
- Agile earned value can use **story points**: planned points versus completed points, and cost per point.
- Only work meeting the definition of done counts as earned.
- Partially done stories earn nothing. There is no '80 percent done'.

> ON THE EXAM
>
> On agile work, 'earned' means done to the definition of done. Burnups show it; module 29 covers them.

Working with governance on variances

- The outline asks you to **monitor financial variations and work with the governance process**.
- Agree **thresholds** in the plan, for example CPI below 0.95 triggers escalation.
- When a threshold is breached: analyse the cause, forecast, prepare options, then escalate.
- Waiting for the numbers to 'recover by themselves' rarely works. Early indices are strong predictors.

> ON THE EXAM
>
> Variances are measured against agreed thresholds. Breach one, and governance hears about it, with options.

Project Lantern: earned value in practice

- Used for the **predictive workstreams**: kiosks, network and migration.
- Progress rule: **0/100** for each clinic's installation, so nothing is 'nearly done'.
- The app uses a **release burnup** instead, in story points.
- Governance threshold: CPI or SPI below 0.92 goes to the steering committee, with options.

> ON THE EXAM
>
> Earned value where the baseline is predictive; burnups where the work is adaptive; thresholds for both.

Scenario signals and the move

THE SCENARIO GIVES	THE MOVE
PV, EV and AC; asks for status	CPI and SPI; below 1 is bad
'Current performance will continue'	EAC = BAC / CPI
'The overrun was a one-off'	EAC = AC + (BAC - EV)
'What efficiency is needed to finish on budget?'	TCPI = (BAC - EV) / (BAC - AC)
SPI near 1 near the end, milestones late	Check the critical path; SPI misleads late on
An index breaches the agreed threshold	Analyse, forecast, options, escalate per governance

ON THE EXAM

Start every calculation with EV. Let the question's wording choose the forecast.

> At a monthly review, a project's planned value is $400,000, its earned value is $360,000, and its actual cost is $450,000.
>
> The sponsor asks for a one-line summary of the project's status.

Which summary is correct?

A Under budget and ahead of schedule.

B Over budget and ahead of schedule.

C Over budget and behind schedule.

D Under budget and behind schedule.

◆ PAUSE AND DECIDE

Which summary is correct?

A Under budget, ahead.

Both indices are below 1. Both parts are wrong.

B Over budget, ahead.

Right about cost; SPI is 0.9, so behind, not ahead.

C Over budget, behind.

CPI = 360 / 450 = 0.8; SPI = 360 / 400 = 0.9. Both below 1.

D Under budget, behind.

Right about schedule; CPI is 0.8, so over budget.

THE SIGNAL

Compute both indices. Below 1 on cost: over budget. Below 1 on schedule: behind.

> A project has a budget at completion of $1,000,000. Its earned value is $400,000 and its actual cost is $500,000.
>
> The team expects current cost performance to continue for the rest of the project.

What is the estimate at completion?

A $1,100,000

B $1,000,000

C $1,150,000

D $1,250,000

◆ PAUSE AND DECIDE

What is the estimate at completion?

A $1,100,000

The atypical formula, for a one-off overrun. The question says it will continue.

B $1,000,000

The original budget. Ignores a 20 percent cost overrun.

C $1,150,000

Not produced by any standard formula here.

D $1,250,000

CPI = 400 / 500 = 0.8; EAC = 1,000,000 / 0.8.

THE SIGNAL

'Current performance will continue' means the typical formula: EAC = BAC / CPI.

> The same project, with a budget at completion of $1,000,000, earned value $400,000 and actual cost $500,000, is reviewed again.
>
> The team now establishes that the entire overrun came from a one-off equipment failure that will not recur. The remaining work should go to plan.

What is the estimate at completion?

A $1,100,000

B $1,000,000

C $1,150,000

D $1,250,000

◆ PAUSE AND DECIDE

What is the estimate at completion?

A $1,100,000

AC + (BAC - EV) = 500,000 + 600,000. The one-off stays; the rest goes to plan.

B $1,000,000

Ignores the $100,000 already lost to the failure.

C $1,150,000

Not produced by any standard formula here.

D $1,250,000

The typical formula. Assumes the poor performance continues, which it will not.

THE SIGNAL

Same numbers, different story, different formula. A one-off overrun: EAC = AC + (BAC - EV).

> A project has a budget at completion of $200,000. Its earned value is $120,000 and its actual cost is $140,000.
>
> The sponsor insists that the project must still finish within the original budget.

What cost performance must the team achieve on the remaining work?

A 0.75

B 1.33

C 0.86

D 1.17

◆ PAUSE AND DECIDE

What cost performance must the team achieve on the remaining work?

A 0.75

The TCPI formula upside down: money remaining over work remaining.

B 1.33

TCPI = (200 - 120) / (200 - 140) = 80 / 60.

C 0.86

The current CPI, 120 / 140. The question asks about the future.

D 1.17

Not produced by the TCPI formula.

THE SIGNAL

TCPI: work remaining over money remaining. Compare it with the current CPI to judge realism.

> Near the end of a construction project, the schedule performance index is 0.98. However, the building's handover milestone, which is on the critical path, is forecast to be three weeks late.
>
> The sponsor asks why the index looks healthy.

What should the project manager do?

A Report the project as on schedule, based on the SPI.

B Rely on the SPI, since it combines all the schedule information.

C Explain that SPI converges towards 1 late in a project, and report schedule status from critical path analysis.

D Rebaseline the schedule so the SPI and the milestone agree.

◆ PAUSE AND DECIDE

What should the project manager do?

A Report on schedule.

Misleading. The critical milestone is three weeks late.

B Rely on SPI.

SPI ignores the critical path and drifts towards 1 near the end.

C Explain the SPI limit; use critical path status.

Honest, accurate status from the right tool.

D Rebaseline to match.

Hides a real delay by moving the target. Not what baselines are for.

THE SIGNAL

Late in a project, SPI flatters. For schedule truth, look at the critical path.

> The governance plan says that a cost performance index below 0.92 must be escalated to the steering committee. This month the CPI has fallen to 0.88.
>
> The project manager believes a recovery is possible.

What should the project manager do?

A Wait a month to see whether performance recovers before reporting.

B Reduce scope quietly to bring costs back into line.

C Rebaseline the cost so the CPI returns to 1.

D Analyse the cause, update the forecast, prepare recovery options, and escalate as the governance plan requires.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait a month.

Breaches the agreed threshold, and early indices rarely recover alone.

B Cut scope quietly.

An unauthorised change that hides the problem from governance.

C Rebaseline.

Hides the variance instead of managing it. Needs change control anyway.

D Analyse, forecast, options, escalate.

Works with the governance process, arriving with analysis and a plan.

THE SIGNAL

A breached threshold goes to governance, promptly, with analysis, a forecast and options. Recovery ideas belong in the options.

Start with EV. **Subtract** for variances, **divide** for indices. Let the story choose the forecast.